Lumora Labs LLM Evaluation Playbook

LLM Evaluation Playbook - version 1.4 - effective 2026-07-01. Owner: Tomasz Wierzbicki, Evaluation Lead.

1. Why we evaluate

LLM features fail quietly: a prompt tweak, a new chunk size or a model upgrade can lower answer quality without raising a single error. Every change to a production LLM feature is therefore measured on a fixed golden set before release, and the results are compared with the current production version.

2. Golden sets

- Each production assistant has a golden set of at least 150 questions.
- At least 15% of the questions must be unanswerable from the corpus, to measure abstention.
- Relevance is labelled by evidence span: the document and the character offsets of the text that supports the answer. Chunk ids are never used as labels, because they change whenever chunking changes.
- Each question also has a reference answer written by a domain expert.
- Golden sets are refreshed every quarter and after every AI-SEV1 or AI-SEV2 incident.

3. Retrieval metrics

A retrieved chunk counts as relevant when its character span overlaps an evidence span of the question. Retrieval metrics are computed on answerable questions only, at a cut-off K equal to the number of passages sent to the generator.

- Precision@K: the fraction of the top K retrieved chunks that are relevant.
- Recall@K: the fraction of the question's evidence spans covered by at least one of the top K chunks.
- Hit rate@K: the fraction of questions with at least one relevant chunk in the top K.
- MRR (mean reciprocal rank): the average over questions of 1 / rank of the first relevant chunk, counting 0 when no relevant chunk is retrieved. MRR rewards putting the right passage first.
- nDCG@K is reported for information but is not a release gate.

4. LLM-as-judge

Answer quality is scored by an LLM judge, a Tier-1 model run at temperature 0 with a fixed rubric. The judge scores three dimensions, each normalised to the range 0 to 1:

- Faithfulness: every claim in the answer is supported by the retrieved passages.
- Answer relevance: the answer addresses the question that was asked.
- Correctness: the answer agrees with the reference answer.

LLM judges have known biases: position bias (preferring the first of two options), verbosity bias (preferring longer answers) and self-preference (rating output from their own model family higher). The team mitigates them by scoring one answer at a time against a rubric instead of comparing pairs, by asking for a short justification before the score, and by calibrating the judge against human labels. A judge prompt may be used only if it agrees with human labels on at least 85% of a 50-question calibration sample.

5. Abstention accuracy

Abstention accuracy is the fraction of questions where the system correctly abstained on an unanswerable question or correctly answered an answerable one. It is reported together with the false-abstention rate (answerable questions the system refused) because a system can trivially avoid hallucinations by refusing everything.

6. Release gates

A release candidate must meet every blocking threshold. In addition, no blocking metric may drop by more than 2 points (0.02) compared with the current production version, even if it is still above its threshold.

Metric	Threshold	Blocking
Recall@5	>= 0.85	Yes
MRR	>= 0.70	Yes
Faithfulness	>= 0.90	Yes
Answer correctness	>= 0.80	Yes
Abstention accuracy	>= 0.90	Yes
Answer relevance	>= 0.85	No (warning)
P95 latency, fast mode	<= 4 seconds	No (warning)
P95 latency, agentic mode	<= 12 seconds	No (warning)
Cost per 1K requests	Within Model Selection Policy cap	Yes

The CI pipeline runs the offline evaluation on every pull request that changes prompts or retrieval configuration and fails the build when a blocking gate fails.

7. Online monitoring

In production, 2% of requests are sampled and scored by the judge for faithfulness. The hallucination rate is one minus the faithfulness pass rate on this sample. A hallucination rate above 8% over a rolling 24-hour window raises an AI-SEV3 incident, and above 15% an AI-SEV2 incident, as described in the LLMOps Incident Runbook.

8. Comparing runs

Every evaluation run stores the full configuration it used (chunking, retrieval, model tiers, prompt versions) together with its metrics, so that any two runs can be compared and any result can be reproduced. A comparison report lists each metric for both runs, the difference, and the questions whose outcome changed.